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Deployment-Oriented Knowledge Distillation for Lightweight Streaming Speech Recognition

Balancing Recognition Accuracy and Deployment Efficiency

HAO WEN

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Supervisor: Haibo Li
Examiner: Anders Hedman
School of Electrical Engineering and Computer Science
Swedish title: Distributionsorienterad Kunskapsdestillering för Lättviktig Strömmande Taligenkänning
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This paper proposes a resource-constrained deployment knowledge distillation framework for lightweight streaming automatic speech recognition (ASR) systems. The proposed framework adopts a teacher–student distillation strategy to transfer semantic representation knowledge from a high-capacity streaming Conformer teacher model to a lightweight streaming CTC-only student model.

Experiments are conducted on the AISHELL-1 Mandarin speech dataset. The results distinguish between the effect of lightweight architectural compression and the additional effect of knowledge distillation. Switching to the lightweight student architecture reduces the parameter size by 81.6% and achieves 3.15× faster inference speed compared with the teacher model. Furthermore, the proposed distillation pipeline helps mitigate the recognition accuracy degradation caused by architectural compression.

From a deployment perspective, the results show a technical trade-off between recognition accuracy and model-side inference efficiency, suggesting the feasibility of lightweight streaming ASR for delay-sensitive deployment scenarios.

SAMMANFATTNING

Denna uppsats presenterar ett ramverk för ”knowledge distillation” (kunskapsöverföring) anpassat för praktisk användning av lättviktsmodeller för automatisk taligenkänning (ASR) i realtid. Ramverket använder en ”lärare-elev”-strategi (teacher-student) för att överföra kunskap från en stor och kraftfull Conformer-modell (läraren) till en mycket mindre och enklare CTC-modell (eleven).

Testerna genomfördes på taldatamängden AISHELL-1. Resultaten skiljer mellan effekten av den lättviktiga modellarkitekturen och den ytterligare effekten av kunskapsdestillering. Genom att använda den mindre elevmodellen minskas antalet parametrar med 81,6%, samtidigt som inferenshastigheten blir 3,15 gånger snabbare jämfört med lärarmodellen. Dessutom bidrar den föreslagna destilleringsmetoden till att minska den försämring i igenkänningsnoggrannhet som orsakas av modellkomprimering.

Ur ett implementeringsperspektiv visar resultaten en teknisk avvägning mellan taligenkänningsnoggrannhet och modellbaserad inferenseffektivitet, vilket tyder på att lättviktiga strömmande ASR-system är en möjlig lösning för tillämpningar med höga krav på låg latens.

CCS Concepts: • **Computing methodologies** → **Speech recognition**; *Neural networks*; *Machine learning*; • **Human-centered computing** → Interaction Format.

Keywords: Streaming Automatic Speech Recognition, Knowledge Distillation, Conformer, CTC-only ASR, Lightweight Speech Recognition, Low-Latency Deployment

Nyckelord: Strömmande automatisk taligenkänning, kunskapsdestillation, Conformer, CTC-baserad ASR, lättviktig taligenkänning, låglatensdriftsättning

Author’s address: Hao Wen, haowe@kth.se, KTH Royal Institute of Technology, School of Electrical Engineering and Computer Science, Stockholm, Sweden, SE 100 44.

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1 INTRODUCTION

1.1 Research Background

Automatic Speech Recognition (ASR) systems have become a core component of modern interactive systems. In real-time interaction scenarios such as voice assistants, in-vehicle systems, and AR/VR, users are typically highly sensitive to system response speed. Research has shown that when speech interaction delay is too high, users are prone to perceiving the system as 'slow' or 'choppy,' which reduces interaction quality and system usability.

In recent years, end-to-end ASR models represented by Transformer and Conformer have achieved significant progress in recognition accuracy. However, such high-capacity models typically have high computational complexity and inference delay, making them difficult to deploy efficiently on edge devices and in low-delay streaming scenarios. Particularly in attention-based decoding formats, models usually rely on full speech context for prediction, thereby further increasing real-time inference costs.

Therefore, in real-time streaming systems, lightweight CTC-based decoding is often adopted to achieve frame-level prediction and low-delay inference. However, compared with encoder–decoder architectures, lightweight CTC-only models lack global semantic modeling capabilities and suffer from a performance decline in complex sound environments.

This reveals the core conflict in current streaming speech recognition systems: systems typically need to balance between recognition performance and low-latency inference efficiency. Therefore, how to maintain high recognition performance while ensuring low-delay streaming inference efficiency has become an important research problem in current streaming ASR systems.

Motivated by low-latency requirements in real-time speech interaction, this thesis treats the problem as a technical deployment study rather than a direct user-experience evaluation. The focus is the trade-off between recognition accuracy and inference efficiency in lightweight streaming ASR.

Specifically, this thesis evaluates how a lightweight CTC-only student architecture reduces model size and inference time, and whether knowledge distillation can mitigate the resulting accuracy degradation. Different distillation strategies, including KL distillation, feature alignment, and hybrid distillation, are implemented and compared.

1.2 Problem Statement

Although existing speech recognition models have made significant progress in recognition accuracy for interactive real-time scenarios, a critical technical conflict remains between complex model architectures and strict low-latency requirements. In streaming speech interaction, excessive inference delay makes even highly accurate models impractical for real-world deployment.

However, while most existing research has primarily focused on pushing the limits of recognition accuracy, there are still limited studies on how lightweight streaming ASR models can effectively balance recognition performance and inference efficiency under real-time deployment limitations.

Therefore, this thesis addresses the following core research question: In streaming ASR systems, how can the technical trade-off between recognition accuracy and inference efficiency be balanced? Specifically, under the premise of using a lightweight student architecture to support low-delay inference, can a knowledge distillation pipeline effectively mitigate the performance decline of the CTC-only model?

1.3 Research Objective

Motivated by the above question, this thesis aims to explore the technical trade-off between recognition accuracy and inference efficiency in streaming speech recognition systems.

On the one hand, the model needs high computational efficiency to support low-delay streaming inference in resource-limited scenarios; on the other hand, the system still needs to maintain enough recognition accuracy for practical deployment.

To this end, this thesis uses a knowledge distillation pipeline as the core approach. It explores how to transfer the semantic modeling capability from a high-capacity Conformer teacher model to a lightweight streaming CTC-only student model. This helps reduce the performance decline caused by the lightweight architecture, while the lightweight student model itself supports low-delay streaming inference.

1.4 Contributions

This thesis makes the following technical contributions:

- **Methodology:** A knowledge distillation pipeline for CTC-only streaming ASR, combining hard-label pretraining with masked KL distillation and feature alignment. The framework decouples training and deployment: it uses a lightweight student architecture to enable low-delay streaming inference, while using the distillation strategy to mitigate the recognition performance decline.
- **Empirical Results:** This thesis quantifies the technical trade-off using CER, parameter size, inference time, and Real-Time Factor (RTF). Switching to the lightweight student architecture achieves an 81.6% parameter reduction (48.76M $\rightarrow$ 8.96M) and 3.15$\times$ faster inference (51.82 ms $\rightarrow$ 16.47 ms) compared to the teacher model. Meanwhile, the distillation pipeline maintains a CER of 9.76% on the AISHELL-1 Mandarin speech corpus. The resulting RTF is reduced to 0.0033, making the model suitable for low-latency deployment scenarios. The reduced computational demand may contribute to lower energy use during deployment, although energy consumption was not directly measured in this work.
- **Systematic Analysis:** Compares different distillation combinations, including hard-label learning, CTC-KL distillation, and feature alignment. The experimental results verify the feasibility of hybrid distillation for lightweight streaming ASR optimization, providing a verified proof of concept for future self-supervised training.

The remainder of this thesis is organized as follows: Section 2 presents the relevant background on ASR, Conformer, CTC, knowledge distillation, and human perception of delay. Section 3 presents the experimental setup and results analysis. Section 4 discusses system deployment and interaction implications. Section 5 concludes the thesis.

2 BACKGROUND

2.1 Automatic Speech Recognition

ASR aims to convert input speech signals into matching text sequences. In recent years, end-to-end ASR models based on deep learning have achieved significant improvements in recognition accuracy and real-time speech processing capabilities [1, 2]. Mainstream approaches mainly include Connectionist Temporal Classification (CTC)-based models and attention-based encoder–decoder models [5, 8, 11, 32].

CTC methods achieve monotonic alignment between input speech and output text through frame alignment prediction, offering inherent low-delay advantages and thus being widely used in streaming speech recognition scenarios [11, 12]. In contrast, attention-based models are capable of modeling global contextual information and typically achieve better recognition performance, but their inference process often relies on higher computational costs and longer context information [6, 8, 32].

In recent years, the Conformer model, which combines self-attention mechanisms with convolutional modules, has further improved ASR system performance by modeling both long-range dependencies and local sound features [13]. Furthermore, large-scale pre-trained models (e.g., Whisper) have further shown that scaling up model size and training data can significantly improve recognition robustness in complex environments [4, 23].

However, in practical streaming deployment scenarios, high-capacity models are often limited by inference delay and computational resources [19]. To meet real-time interaction requirements, streaming ASR systems typically adopt chunk-based streaming or streaming architectures that limit access to future context [17, 18, 29, 33]. Although these methods can reduce inference delay, limited context information may also lead to a decline in model recognition performance.

Overall, while most existing ASR research focuses primarily on pushing the limits of recognition accuracy, there remains a critical need to address the technical trade-off between recognition performance and inference efficiency under the strict low-delay limits of real-time interaction scenarios.

2.2 Conformer Architecture

Conformer is one of the most widely used backbone architectures in modern end-to-end ASR systems. Its key idea is to combine self-attention for long-range context modeling with convolution for local acoustic feature extraction. [13]. Compared with the traditional Transformer architecture, Conformer further integrates convolutional modules with self-attention mechanisms, thereby capturing both long-range contextual information and local sound features, achieving superior performance in speech recognition tasks.

Among them, the self-attention mechanism is capable of establishing global dependencies across different time steps [32], and further enhances sequence modeling capability when combined with relative position encoding [10, 28]. Meanwhile, the convolutional module effectively models the local continuous structure in speech signals, which helps balance out the self-attention mechanism's limitations with local sound details. Furthermore, Conformer employs a Macaron-style feed-forward network to enhance feature representation quality and training stability [20].

However, Conformer has a strong dependence on global contextual information, and the computational complexity of the self-attention mechanism increases significantly with sequence length [32]. Therefore, in real-time streaming speech recognition scenarios, Conformer typically needs to be deployed with chunk-based streaming or streaming strategies that limit access to future context [29].

Although streaming architectures can effectively reduce inference delay, limited context information may also lead to a decline in model recognition performance. This characteristic of "high performance but high computational cost" means that deploying Conformer in real-time interaction scenarios still faces significant challenges, and has further driven the development of lightweight model design and knowledge distillation methods [24].

2.3 CTC-based Streaming ASR

Connectionist Temporal Classification (CTC) is a sequence modeling method widely used in end-to-end speech recognition [11, 12]. CTC achieves monotonic alignment between input speech and output text through frame alignment prediction, offering inherent low-delay advantages and being particularly well-suited for streaming speech recognition scenarios [11, 14].

Compared with attention-based encoder–decoder architectures, the CTC inference process does not rely on a token-by-token attention decoder, thereby effectively reducing model inference complexity and deployment costs [8, 33]. However, due to the lack of explicit global semantic context modeling capability, lightweight CTC-only models typically suffer from a performance decline in complex sound environments.

Therefore, how to further improve model recognition performance while maintaining CTC's low-delay streaming inference efficiency has become an important research problem in current lightweight streaming ASR systems [13, 29].

2.4 Knowledge Distillation

In the standard distillation framework, the student model not only learns from ground-truth labels but is also trained by minimizing the difference between its output distribution and that of the teacher model [15, 25]. To this end, knowledge distillation typically introduces softmax with temperature scaling and KL loss [15].

The standard softmax function can be expressed as:

$$p_i = \frac{e^{z_i}}{\sum_j e^{z_j}} \tag{1}$$

The softmax with temperature scaling is:

$$p_i(T) = \frac{e^{z_i/T}}{\sum_j e^{z_j/T}} \tag{2}$$

where a higher temperature parameter produces a smoother output distribution, which helps improve richer inter-class relationship information. The distillation loss is typically defined as [15]:

$$\mathcal{L}_{KD} = \mathrm{KL}\left(P_T \parallel P_S\right) \tag{3}$$

where P_T and P_S represent the temperature-scaled output probability distributions of the teacher model and the student model.

In addition to output-level distillation, researchers have further proposed distillation methods based on intermediate-layer feature alignment, such as FitNet [25]. This approach further improves lightweight models by aligning the hidden feature representations of the teacher model and the student model. The feature distillation loss is typically expressed as:

$$\mathcal{L}_{feat} = \|H_T - H_S\|_2^2 \tag{4}$$

where H_T and H_S represent the hidden feature representations of the teacher model and the student model at matching layers.

In the field of speech recognition, knowledge distillation has been widely applied to model compression and deployment optimization tasks [9, 16, 30, 31]. Particularly in CTC-based or attention-based

end-to-end systems, the teacher model can guide the student model to learn more stable alignment patterns and semantic representations, which helps improve the robustness and recognition performance of lightweight models in complex sound environments [15, 16].

However, most existing knowledge distillation methods primarily focus on improving recognition accuracy in offline evaluations, with limited systematic optimization for the strict inference delay and deployment limitations required in real-time streaming scenarios [3, 27].

2.5 Human Perception of delay

In human-computer interaction systems, the actual delay of the system and the delay perceived by the user are not always consistent. Existing research has shown that users' evaluation of system responsiveness depends not only on the actual time cost but is also influenced by factors such as task type, cognitive load, and interaction context [21, 22, 27].

This difference is particularly evident in speech interaction scenarios. When system response delay is low, users typically perceive it as part of a natural conversation. However, when delay reaches the order of several hundred milliseconds, users become distinctly aware of an interruption in the interaction rhythm, leading to subjective feelings that the system is "slow" or "choppy" [3, 26].

As a typical delay-sensitive interface, speech interaction systems are highly sensitive to response speed. In continuous dialogue and real-time control scenarios, users typically expect the system to complete understanding and feedback quickly; otherwise, the coherence and immersion of the interaction are disrupted [26, 27].

Therefore, in the design of streaming speech recognition systems, the optimization objective should not be limited solely to recognition accuracy, but also needs to comprehensively consider the impact of system response delay on user interaction experience. This shift from "model performance optimization" to "human-perception-driven optimization" provides an important motivation for the method design in the following sections of this paper.

These perception studies motivate the use of computational latency metrics in this thesis. Although subjective user experience is not directly evaluated, recognition accuracy, inference time, and RTF are used as measurable proxies for deployment suitability in delay-sensitive speech interaction systems.

2.6 System Overview

This section presents the overall design of the proposed streaming speech recognition system, as shown in Figure 1.

The system adopts a teacher–student knowledge distillation framework [15] combined with a CTC-only streaming deployment strategy, aiming to balance recognition performance and system inference latency in real-time interaction scenarios.

During the training phase, a high-capacity Conformer teacher model [13] is first constructed to learn rich semantic information and temporal alignment relationships. Subsequently, a lightweight streaming student model is introduced, and the teacher model's global modeling capability is transferred to the student through knowledge distillation.

The system adopts a training–inference decoupling design strategy. During the training process, both the teacher and student models are involved in the optimization, while in the actual deployment phase, only the lightweight student model is retained for streaming inference.

Because the student model utilizes a native CTC-only architecture, it can directly perform frame alignment prediction without relying on an autoregressive attention decoder during inference, thereby effectively reducing model complexity and system response latency.

This design reflects a technical trade-off between recognition performance and model-side inference latency, providing a foundation for system-level responsiveness. During the training phase, the high-capacity teacher model provides richer global semantic information to enhance the student's representation quality. In the deployment phase, the lightweight CTC-only structure significantly reduces model parameters and inference costs, fulfilling the strict low-latency requirements of real-time streaming interaction.

From a system deployment perspective, this design serves as a proof of concept, demonstrating the feasibility of mitigating the performance decline caused by lightweight architectures while ensuring low-latency streaming inference.

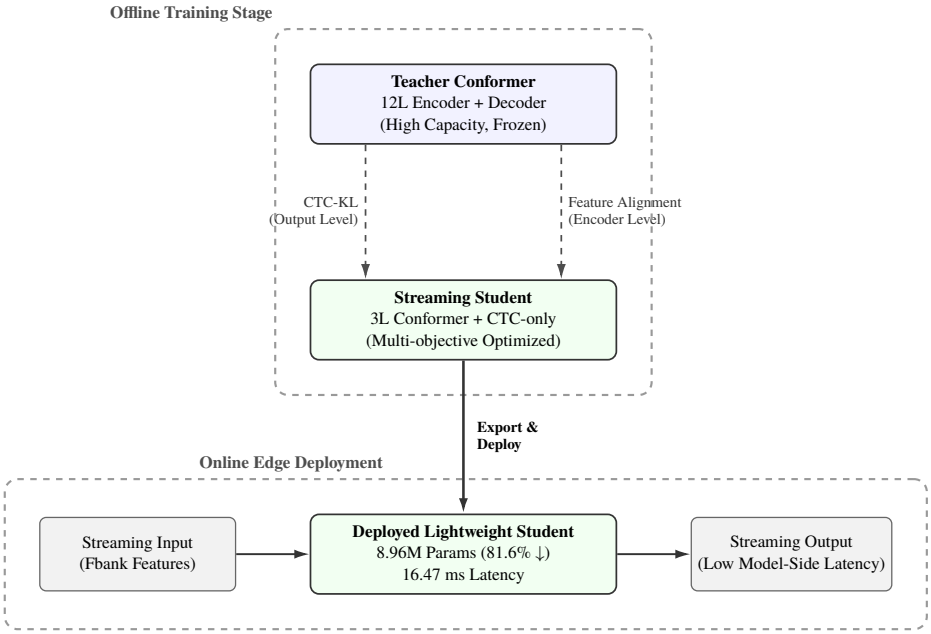


Figure 1: Overview of the proposed lightweight streaming ASR framework. During the offline training stage (top), a high-capacity Conformer teacher transfers knowledge to a CTC-only student. During the online stage (bottom), the exported student model processes streaming inputs with low model-side inference latency for delay-sensitive streaming applications.

2.7 Teacher and Student Architectures

2.7.1 Teacher Architecture. The teacher model adopts a high-capacity streaming Conformer architecture, consisting of a 12-layer Conformer encoder and a 6-layer Transformer decoder. The encoder is responsible for the temporal modeling of input speech features and contextual information extraction, serving as the acoustic modeling component. Meanwhile, the decoder utilizes attention mechanisms to learn stronger semantic dependencies, serving as the language modeling component.

With its higher parameter capacity and stronger global contextual modeling capabilities, the teacher model can learn richer semantic representations and temporal alignment information. This enables it to provide high-quality supervision signals for the subsequent knowledge distillation pipeline.

To make sure the training phase matches real-time streaming deployment scenarios, the teacher model incorporates causal attention and dynamic chunking mechanisms.

2.7.2 Student Architecture. The student model uses a lightweight streaming Conformer-CTC architecture, consisting of only a 3-layer Conformer encoder and a CTC output layer, without an attention decoder.

Compared to the teacher model, the student model has fewer encoder layers and a lighter network structure. This directly reduces the parameter size and inference complexity.

Like the teacher, the student model also uses causal attention and dynamic chunking mechanisms to meet the low-latency requirements of real-time streaming speech recognition scenarios.

Because it uses a native CTC-only architecture, the student model can perform frame-level prediction directly without relying on token-by-token decoding. This design reduces inference latency and improves computational efficiency.

As a result, the student model is suitable for edge deployment on edge devices, real-time voice assistants, and delay-sensitive speech applications.

2.8 Distillation Objectives

To reduce the performance decline caused by the lightweight CTC-only architecture, this thesis uses a multi-objective knowledge distillation strategy to optimize the student model. The overall loss function is defined as follows:

$$L = \lambda_1 L_{CTC} + \lambda_2 L_{KL} + \lambda_3 L_{feat} \tag{5}$$

where L_{CTC} is the hard-label CTC loss of the student model; L_{KL} is the KL distillation loss between the CTC output distributions of the teacher and student models; L_{feat} is the feature alignment loss between their encoder hidden representations; and λ_1, λ_2, and λ_3 are the weight parameters for each loss term.

2.8.1 CTC Loss. The CTC loss helps the student model learn the speech-text alignment from the ground-truth labels. Because CTC performs sequence alignment without requiring explicit frame-level labels, it is highly suitable for streaming ASR.

Through this loss, the student model acquires basic recognition capabilities and establishes frame-level prediction.

2.8.2 CTC-KL Distillation. To transfer the teacher's global semantic modeling capability to the lightweight student, this thesis adds KL distillation based on the CTC output distributions.

First, temperature scaling is applied to the logits of both models to compute their probability distributions:

$$P_s = \text{softmax}\left(\frac{z_s}{T}\right), \quad P_t = \text{softmax}\left(\frac{z_t}{T}\right) \tag{6}$$

where z_s and z_t are the CTC logits of the student and teacher models, and T is the temperature parameter.

Next, the KL divergence minimizes the difference between these output distributions at each time step t:

$$L_{KL}(t) = \sum_{c} P_t(c|x_t) \log \frac{P_t(c|x_t)}{P_s(c|x_t)} \tag{7}$$

where c is the vocabulary class, and x_t is the input at time step t.

To reduce the impact of low-confidence outputs and blank tokens on training stability, this thesis uses a high-confidence non-blank mask. This limits distillation only to time steps where the teacher model has high-confidence and non-blank predictions.

Specifically, the distillation mask is defined as:

$$M_t = \mathbb{1}(\max(P_t^{raw}) > \tau \wedge y_t \neq blank) \tag{8}$$

where τ represents the confidence threshold, and P_t^{raw} is the teacher's output probability distribution without temperature scaling.

Finally, the Masked KL Distillation Loss is defined as:

$$L_{KL}^{masked} = T^2 \cdot \frac{\sum_t M_t \cdot L_{KL}(t)}{\sum_t M_t} \tag{9}$$

where T^2 compensates for the gradient scaling effect introduced by the temperature.

2.8.3 Feature Alignment. In addition to output-level distillation, this paper further introduces encoder hidden feature alignment.

To help the student model learn more stable intermediate representations from the teacher model, the hidden features of the teacher model and the student model are further aligned.

The hidden representations of the student model and the teacher model are defined respectively as:

$$H_s^i, \quad H_t^i$$

where i represents the hidden layer index.

Since the teacher model and the student model use the same hidden feature dimension, hidden feature alignment can be directly performed between matching encoder layers.

Next, Mean Squared Error (MSE) is used to reduce the difference between the hidden representations of the teacher model and the student model:

$$L_{feat} = \frac{1}{N} \sum_{t} \left\| H_s^i(t) - H_t^i(t) \right\|_2^2 \tag{10}$$

where N represents the number of valid time steps.

To improve distillation stability, this paper only aligns the final hidden representations of the teacher model and the student model. Since the student model has already been pretrained, this design further focuses on transferring high-level semantic representations while reducing the influence of multi-layer feature distillation on the training stability of the lightweight streaming architecture.

2.9 Training Strategy

Due to the limited model capacity of the lightweight CTC-only architecture and the lack of additional language modeling provided by an attention decoder, directly performing multi-objective knowledge distillation from random initialization may lead to unstable optimization during the early training stages.

To improve training stability, this thesis uses a two-stage training strategy. As shown in Figure 2, the framework first pre-trains the student using pure CTC, followed by multi-objective distillation.

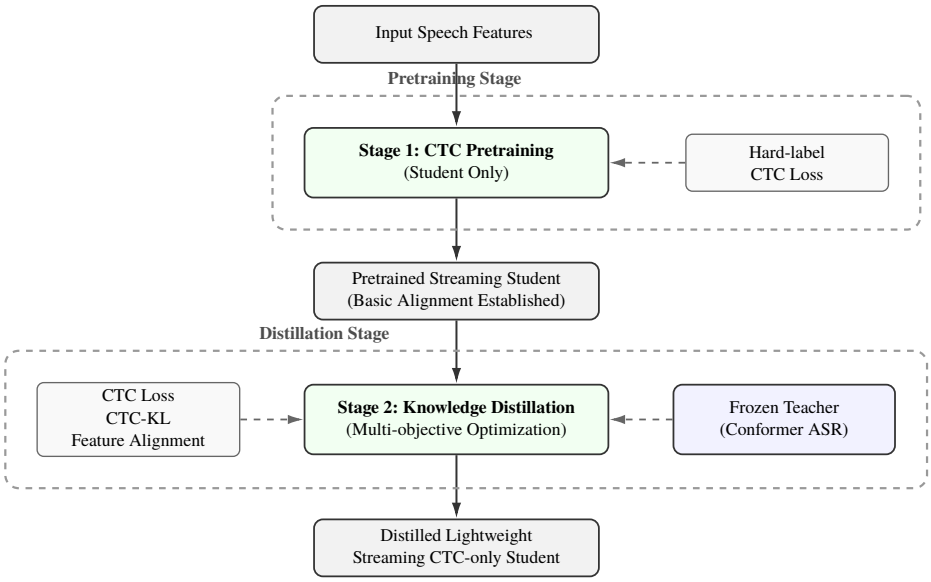


Figure 2: Two-stage training strategy for lightweight streaming CTC-only ASR distillation.

In the first stage, the student is pre-trained using only the hard-label CTC loss. This establishes basic recognition capabilities and initial speech-text alignment.

In the second stage, the pre-trained student undergoes multi-objective knowledge distillation, incorporating both CTC-KL output distribution and encoder feature alignment.

Compared to distilling directly from random initialization, this two-stage approach effectively stabilizes early training. It also reduces the optimization difficulties typically caused by combining a lightweight model with a limited-context streaming architecture.

During the distillation process, the teacher's parameters remain frozen; only the student's parameters and the hidden feature alignment layer are optimized.

This training strategy allows the CTC-only student to stably learn the teacher's high-quality semantic representations while maintaining its own low model complexity and real-time inference efficiency.

As shown in subsequent experiments, this distillation pipeline effectively mitigates the performance decline caused by the lightweight streaming architecture. Therefore, the proposed edge-oriented framework serves as a practical proof of concept, demonstrating a technical trade-off between recognition accuracy, deployment complexity, and model-side inference speed.

3 EXPERIMENTS AND ANALYSIS

3.1 Experimental Setup

All models are trained and evaluated on the AISHELL-1 Mandarin speech dataset [7]. The input consists of 80-dimensional Fbank features with a 16 kHz sampling rate, normalized using global Cepstral Mean and Variance Normalization (CMVN).

The teacher uses a high-capacity streaming Conformer, while the student uses a lightweight Conformer-CTC architecture. Both incorporate causal attention and dynamic chunking mechanisms to meet the low-latency requirements of streaming ASR scenarios.

The system is implemented using PyTorch and the WeNet toolkit [33, 34], with all training conducted on NVIDIA A800 GPUs.

The experiments are evaluated using the following core metrics:

- Character Error Rate (CER)
- Model parameter size (Complexity)
- Inference latency
- Real-Time Factor (RTF)

Training uses the Adam optimizer with a learning rate of 0.001. Following the two-stage strategy, the student is first pre-trained with pure CTC and then optimized via multi-objective distillation. The loss weights are set to $\lambda_1 = 1.0$, $\lambda_2 = 0.2$, and $\lambda_3 = 0.2$, with a KL distillation temperature of $T = 3.0$.

Because this thesis focuses on real-time deployment streaming ASR, the experiments primarily validate the feasibility of the distillation pipeline on lightweight CTC-only models, rather than pursuing state-of-the-art (SOTA) recognition accuracy.

Detailed training configurations for the teacher model, student model, and distillation settings are provided in the Appendix.

3.2 System Performance Results

Table 1 presents the performance of the CTC-only student model under different distillation strategies.

Table 1: Performance comparison under different distillation strategies.

Model	Strategy	$\text{CER}_{\text{Greedy}}$	CER_{Beam}
Baseline Student	None	10.18	10.17
Post-trained Student	Hard-label only	9.94	9.94
KL Distillation	KL = 0.2	9.88	9.83
Feature Distillation	MSE = 0.2	9.90	9.90
Hybrid Distillation	KL = 0.2 + MSE = 0.2	9.80	9.76

The experimental results indicate that knowledge distillation effectively improves the student's recognition accuracy. While continuous training with only hard labels yields baseline improvements, introducing KL distillation or feature alignment further mitigates the performance decline caused by the lightweight architecture.

Notably, hybrid distillation (KL + Feature Alignment) achieves the lowest Character Error Rate (CER). This shows that output distribution matching and intermediate feature alignment provide complementary supervision signals for the lightweight streaming model.

Overall, these findings validate the core feasibility of this approach: multi-objective distillation successfully mitigates accuracy degradation even under an extremely lightweight streaming architecture. The main deployment efficiency gain comes from the lightweight CTC-only student architecture, while knowledge distillation provides a smaller but consistent improvement in recognition accuracy.

3.3 Ablation Study

To isolate and analyze the contribution of each distillation component, this thesis conducts ablation experiments.

Table 2: Ablation study of different distillation components.

CTC	KL	Feature MSE	CER
✓	–	–	9.94
✓	✓	–	9.83
✓	–	✓	9.90
✓	✓	✓	9.76

3.3.1 Distillation Loss Components. As shown in Table 2, relying solely on the hard-label CTC loss establishes a baseline CER of 9.94%. Introducing either KL distillation or feature alignment individually improves the recognition accuracy. Ultimately, the model achieves its best performance (9.76% CER) when both distillation objectives are combined.

The results indicate that KL distillation primarily provides supervision at the output distribution level, while Feature MSE stabilizes intermediate representation learning. By combining both methods, the lightweight streaming student model successfully leverages these complementary supervision signals to maximize the effectiveness of the distillation pipeline.

The limited gain from feature alignment alone suggests that intermediate representation matching is harder to optimize in a compact streaming student than output-level supervision. In contrast, KL distillation directly regularizes the CTC output distribution and therefore provides a more stable learning signal. The best result is obtained when both objectives are combined, indicating that output-level and feature-level supervision are complementary rather than redundant.

3.3.2 Temperature Analysis. To regulate the softening effect of the teacher's output distribution during KL distillation, the temperature parameter is set to $T = 3.0$. Specifically, this thesis monitors the average Top-1 confidence of the scaled teacher outputs on valid non-blank frames. Empirical observations show that after temperature scaling, this value stabilizes between 0.58 and 0.67. This indicates that the distribution becomes adequately smoothed while preserving distinct primary predictions.

Consequently, $T = 3.0$ achieves a practical balance between providing rich inter-class information and maintaining clear teacher guidance, which effectively improves the training stability of the lightweight student model.

3.3.3 Feature Alignment Layers. Preliminary experiments reveal that mismatched hidden dimensions between the teacher and student encoders often lead to unstable training during feature alignment. While a linear projection layer could map these representations into a shared space, it introduces unnecessary optimization complexity for lightweight streaming ASR architectures.

To address this, the output dimension of the student encoder is set to strictly match the teacher encoder (256 dimensions). This design enables direct, projection-free hidden feature alignment. It significantly simplifies the feature-level supervision pipeline and improves distillation stability.

As indicated by the experimental results, relying solely on feature distillation yields limited performance gains. However, when combined with KL distillation, the model achieves its optimal accuracy.

This confirms that while KL distillation establishes stable output-level patterns, direct and projection-free intermediate feature matching provides the necessary structural guidance to learn high-quality intermediate representations.

3.4 Deployment and Interaction Analysis

While this thesis focuses on technical optimization rather than subjective user testing, the proposed system is explicitly designed to meet the rigorous timing requirements of real-time interaction. Therefore, this section primarily evaluates response speed and deployment efficiency during streaming inference.

Compared to traditional encoder-decoder architectures, the proposed lightweight student model does not rely on an attention decoder during inference, significantly reducing model complexity and decoding dependency.

To evaluate deployment efficiency, this thesis compares the structural complexity and inference latency between the teacher model and the final distilled student model. Table 3 shows the architecture breakdown, and Table 4 presents the performance comparison.

Table 3: Model Architecture Comparison

Model	Encoder Layers	Decoder	Parameters
Teacher	12	Transformer Decoder	48.76 M
Distilled Student (KL + MSE)	3	None (CTC-only)	8.96 M

Table 4: Performance Comparison

Model	Inference Time	CER
Teacher	51.82 ms	6.12
Distilled Student (KL + MSE)	16.47 ms	9.76

The teacher uses a 12-layer streaming Conformer encoder and a Transformer decoder. This high capacity provides strong global semantic modeling and rich supervision signals for distillation.

Notably, the teacher model was only trained for 87 epochs. Once it could provide stable and effective distillation supervision, training was halted early. This is because the core objective of this work is to validate the effectiveness of the distillation pipeline on lightweight CTC-only models, rather than pursuing state-of-the-art recognition accuracy with the teacher.

The experimental results indicate that while the teacher achieves a significantly lower CER, it incurs a substantial deployment cost. It contains 48.76M parameters, occupies approximately 185.99MB of static memory, and requires 51.82ms per inference step.

In contrast, the distilled student eliminates the attention decoder and retains only a 3-layer streaming Conformer encoder with a CTC output structure. This allows direct frame-level prediction without autoregressive token-by-token decoding.

Statistically, the distilled student contains only 8.96M parameters (an 81.6% reduction compared to the teacher) and its static memory usage drops to 34.18MB. The inference time is reduced to 16.47ms—achieving a 3.15$\times$ speedup over the teacher—with the RTF dropping to 0.0033. In practical edge deployment, combining this architecture with techniques like model quantization could further reduce storage and latency overheads on resource-constrained devices.

Furthermore, the reported inference time specifically measures the main neural network components, excluding the minor overheads of front-end Fbank extraction and final CTC decoding. This isolates and directly reflects the computational complexity of the core architectures during streaming inference.

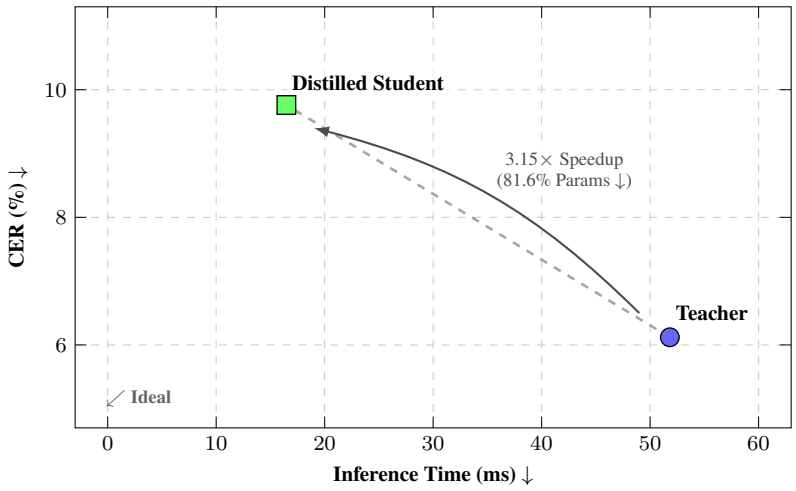


Figure 3: Trade-off between recognition performance (CER) and deployment efficiency measured by model-side inference time. The transition highlights the architectural shift toward low-latency streaming deployment.

As visualized in Figure 3, the distilled student substantially reduces inference latency and deployment complexity while maintaining a stable, practical recognition baseline. From a system design perspective, maintaining low model-side inference latency provides a technical basis for building responsive speech interaction systems, although full interaction quality depends on end-to-end system evaluation.

4 DISCUSSION

4.1 Accuracy and Deployment Trade-off

Although the distilled student model exhibits a higher Character Error Rate (CER) than the teacher model, its substantial reduction in inference latency and computational complexity is beneficial for practical real-time edge applications, which are strictly bound by latency constraints.

As established in system design principles for streaming interfaces, low-latency thresholds act as a hard constraint for continuous interaction. In some interactive systems, downstream dialogue modules may partially tolerate minor recognition errors, but accumulated latency directly affects the continuity of interaction. In contrast, streaming ASR systems cannot easily recover from accumulated inference latency without fundamentally disrupting the real-time interaction pipeline.

Therefore, the primary objective of this thesis is to validate the feasibility of applying knowledge distillation to lightweight streaming CTC-only models. The proposed streaming deployment framework prioritizes establishing a technical trade-off between recognition performance, deployment complexity, and model-side inference efficiency, rather than solely optimizing for the lowest possible CER.

The experimental validation confirms that the lightweight CTC-only streaming student minimizes inference costs while retaining stable, usable recognition performance. This trade-off is fundamental for

deploying AI-native voice applications on devices with limited computational resources and low-latency budgets.

Furthermore, the substantial reduction in structural complexity and inference time reduces computational demand during continuous streaming. By minimizing computational resource demands, this deployment-oriented architecture may contribute to more efficient deployment and aligns with the goals of sustainable computing and Green AI. It demonstrates that optimizing for edge deployment can reduce computational overhead and may contribute to more efficient AI deployment, although energy consumption was not directly evaluated.

4.2 Effectiveness of Knowledge Distillation

The experimental results show that knowledge distillation effectively mitigates the performance degradation caused by the structural constraints of lightweight streaming architectures.

Among all experimental settings, hybrid distillation achieves the best overall performance. This confirms that output distribution matching and intermediate feature alignment provide complementary supervision signals for the student model.

Specifically, CTC-KL distillation helps the student learn stable output patterns, while feature alignment directly guides intermediate semantic representation. Together, these two levels of supervision target different aspects of knowledge transfer, improving training stability of the lightweight streaming ASR model.

4.3 Limitations of Lightweight Streaming ASR

Compared to traditional encoder-decoder architectures, the lightweight CTC-only design inevitably gives up some global contextual modeling in exchange for lower deployment costs and faster inference.

Because the student removes the token-by-token attention decoder, it relies entirely on frame-level CTC prediction. While this design significantly improves deployment efficiency and responsiveness, it may limit semantic modeling in acoustically complex or long-context scenarios.

Moreover, to focus on validating the distillation pipeline efficiently, the teacher model was trained for only 87 epochs rather than to full convergence. Therefore, the student's final performance is bounded by the teacher's recognition limits.

Future work could introduce self-supervised pre-training to improve the teacher's acoustic and semantic representations, providing richer supervision signals for the student. Additionally, structured pruning and model quantization could be explored to further reduce deployment complexity and inference costs.

Finally, because this thesis focuses on deployment-oriented ASR optimization, it does not include formal user studies. Future research could investigate user-centered interaction evaluation in real-time streaming scenarios.

Another limitation is that the deployment evaluation mainly measures neural network inference time, rather than full end-to-end latency on real edge hardware. Front-end feature extraction, decoding overhead, memory access, and device-specific runtime behavior may further affect real deployment performance. Therefore, the reported latency should be interpreted as a controlled comparison of model-side efficiency rather than a complete product-level benchmark.

Therefore, the results in this thesis should be interpreted as an exploratory proof of concept rather than a confirmatory study. The findings show what is possible and identify trade-offs, but further validation with larger-scale experiments and real-world deployment data is needed.

4.4 Implications for Low-Latency Streaming Deployment

From a system design perspective, low-latency speech feedback is a strict hardware requirement for continuous real-time applications.

In streaming scenarios, high processing delays can cause downstream pipeline bottlenecks, breaking the continuity of voice interfaces. Therefore, reducing response delay is a core requirement for system stability, rather than just a basic model compression task.

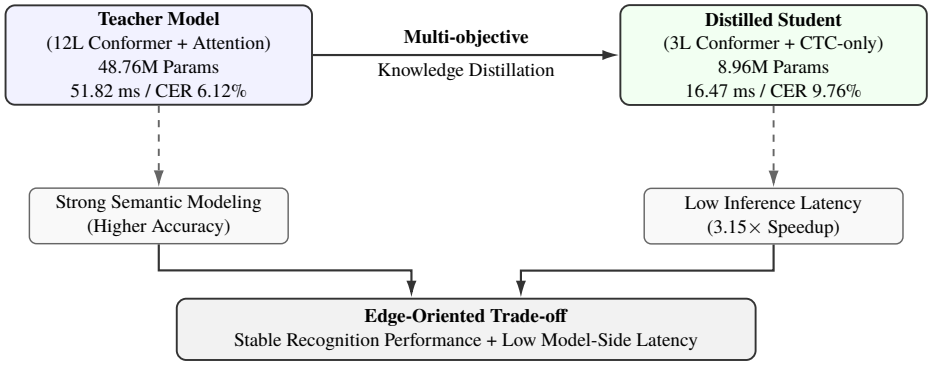


Figure 4: Resource-constrained deployment trade-off between the high-capacity teacher model and the distilled lightweight streaming CTC-only student model.

This is especially important for edge applications such as voice assistants, AR/VR headsets, and in-vehicle systems. In these environments, low-latency budgets are required to maintain system synchrony.

The real-time deployment framework proposed in this thesis shows that improving model-side inference efficiency provides a technical basis for building more responsive systems, although full responsiveness and reliability require end-to-end system evaluation and user testing.

Therefore, lightweight streaming ASR optimization should be viewed not only as a parameter reduction problem, but as an important design factor supporting low-latency speech processing. The impact on end-to-end responsiveness and reliability remains a topic for future evaluation.

4.5 Ethical Considerations

Although this thesis mainly focuses on deployment efficiency and streaming ASR optimization, lightweight speech recognition systems still raise some ethical concerns related to privacy and fairness.

From a deployment perspective, on-device inference can improve privacy because speech data does not need to be sent to cloud servers. This is important for edge devices and real-time speech applications.

However, lightweight models may show different recognition performance for different accents or speaking styles. Therefore, future work should further explore training methods that improve fairness and robustness while still keeping low inference cost and deployment efficiency.

4.6 Sustainability Implications

While this thesis focuses on technical optimization for low-latency streaming ASR, the proposed lightweight architecture has potential sustainability benefits. By reducing the model size from 48.76M to 8.96M parameters (81.6% reduction) and achieving 3.15× faster inference, the system requires less computational resources during continuous deployment. This reduction in computational demand can

translate to lower energy consumption and reduced carbon footprint when deployed at scale on edge devices.

From a Green AI perspective, deployment-oriented model compression matches with the goal of minimizing environmental impact without making performance much worse. The reduced memory footprint (185.99MB $\rightarrow$ 34.18MB) also means lower hardware requirements, potentially extending the lifespan of edge devices and reducing electronic waste.

However, this thesis did not directly measure energy consumption or carbon dioxide output. Future work should include real energy measurements to quantify the environmental benefits of lightweight streaming ASR deployment.

5 CONCLUSION

This thesis proposes and validates a deployment-oriented lightweight streaming ASR framework. By leveraging knowledge distillation, the framework explores the technical trade-off between recognition performance and deployment efficiency for edge-based speech applications.

A decoupled two-stage training strategy is introduced, comprising CTC pretraining and multi-objective distillation via KL divergence and feature alignment. This architecture separates the training capacity from deployment constraints, enabling lightweight streaming inference while effectively inheriting rich global semantic representations from the teacher model.

Experimental results on the AISHELL-1 dataset demonstrate that, compared to the high-capacity teacher, the distilled lightweight student reduces the parameter footprint by 81.6% and achieves a 3.15$\times$ inference speedup while maintaining stable recognition performance. Crucially, the RTF is reduced to 0.0033, supporting the structural feasibility of deploying this CTC-only architecture in low-latency streaming scenarios.

From a system design perspective, this work highlights that deployment efficiency and inference latency provide an important technical foundation for responsive speech interaction systems. Therefore, lightweight streaming ASR optimization must be treated not merely as a parameter reduction task, but as a critical engineering requirement for supporting low-latency model-side processing in streaming deployment scenarios.

Future work can integrate self-supervised pre-training, structured pruning, and model quantization to further optimize deployment costs. Additionally, system-level end-to-end evaluation can be explored to measure interaction reliability in complex acoustic environments. Figure 5 illustrates a comprehensive future optimization pipeline for edge-oriented ASR systems.

Overall, this thesis demonstrates that deployment-oriented knowledge distillation provides a feasible path toward lightweight streaming ASR systems. By reducing inference latency and computational overhead, the proposed framework supports the feasibility of low-latency model-side streaming ASR. However, full real-time interaction quality remains to be validated through future end-to-end evaluations and user-centered studies.

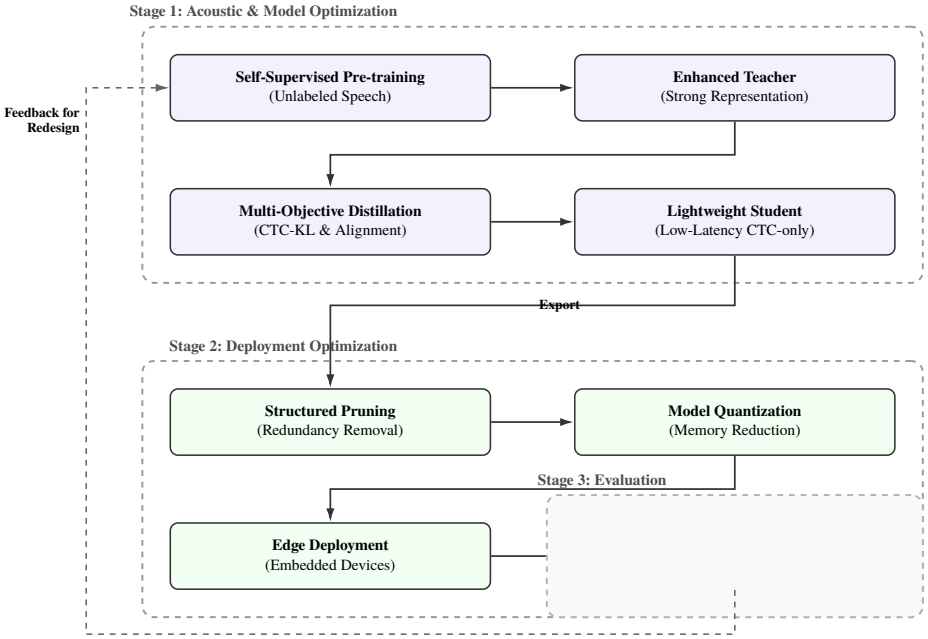


Figure 5: Future streaming deployment MLOps pipeline for lightweight streaming ASR systems. The workflow is organized in a top-down cascaded structure, moving from acoustic optimization to edge deployment, and forming a complete closed loop through system-level evaluation.

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A EXPERIMENTAL DETAILS

This appendix provides the main training configurations of the teacher model, student model, and distillation settings used in the experiments.

A.1 Architecture Summary

A.2 Student Model Training Configuration

```
encoder: conformer

encoder_conf:
    output_size: 256
    attention_heads: 4
    linear_units: 1024
```

Table 5: Architecture comparison between teacher and student models.

Component	Teacher	Student
Encoder layers	12	3
Encoder output size	256	256
Attention heads	4	4
Feed-forward dimension	1024	1024
Decoder	Transformer (6 layers)	None (CTC-only)
Parameters	48.76 M	8.96 M

```
    num_blocks: 3
    dropout_rate: 0.1
    positional_dropout_rate: 0.1
    attention_dropout_rate: 0.0
    input_layer: conv2d
    normalize_before: true
    cnn_module_kernel: 15
    use_cnn_module: true
    activation_type: swish
    pos_enc_layer_type: rel_pos
    selfattention_layer_type: rel_selfattn
    causal: true
    use_dynamic_chunk: true
    use_dynamic_left_chunk: false
    cnn_module_norm: layer_norm
decoder: null
model: asr_model_ctc
dataset_conf:
    fbank_conf:
        num_mel_bins: 80
        frame_shift: 10
        frame_length: 25
    spec_aug: true
    batch_conf:
        batch_type: static
        batch_size: 64
grad_clip: 5
accum_grad: 1
max_epoch: 80
optim: adam
optim_conf:
    lr: 0.001
scheduler: warmuplr
scheduler_conf:
```

```
    warmup_steps: 5000
```

A.3 Teacher Model Configuration

```
encoder: conformer

encoder_conf:
    output_size: 256
    attention_heads: 4
    linear_units: 1024
    num_blocks: 12
    dropout_rate: 0.1
    cnn_module_kernel: 15
    use_cnn_module: true
    pos_enc_layer_type: rel_pos
    selfattention_layer_type: rel_selfattn
    causal: true
    use_dynamic_chunk: true
    use_dynamic_left_chunk: false
decoder: transformer
decoder_conf:
    attention_heads: 4
    linear_units: 1024
    num_blocks: 6
model: asr_model
model_conf:
    ctc_weight: 0.3
    lsm_weight: 0.1
batch_conf:
    batch_size: 8
max_epoch: 87
optim: adam
optim_conf:
    lr: 0.001
scheduler: warmuplr
scheduler_conf:
    warmup_steps: 25000
```

A.4 Distillation Settings

The distillation experiments share the following hyperparameter settings:

```
lambda_ctc: 1.0
lambda_kl: 0.2
lambda_feat: 0.2
temperature: 3.0
```

Table 6: Distillation configurations used in the experiments.

Method	CTC Loss	KL Loss	Feature Alignment
Pure CTC Fine-tuning	✓	–	–
CTC-KL Distillation	✓	✓	–
CTC + Feature Alignment	✓	–	✓
Hybrid Distillation	✓	✓	✓

```
masked_kl: true
non_blank_only: true
confidence_threshold: 0.7
distillation_training:
    max_epoch: 20
optim_conf:
    lr: 0.0005
scheduler_conf:
    warmup_steps: 1000
```

A.5 Training Resources

- Hardware: NVIDIA A800 GPU
- Framework: PyTorch + WeNet
- Teacher model training: 87 epochs (stopped early once stable distillation supervision was achieved)
- Student model training: 80 epochs (Stage 1) + 20 epochs (Stage 2)

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